

module through heat dissipation and converts it into electricity. The hot-water module cools the PV module, thereby increasing its efficiency. It also provides a temperature gradient for the thermoelectric module to operate. In addition, it supplies hot water for domestic and other uses by recovering the waste heat.

Typically, the energy payback time (EPBT) for a solar thermal system is less than for PV systems. The EPBT of a PV system can be reduced by using it in a hybrid system integrating PV with solar thermal components, such as hot-water tubes and thermoelectric generators. This approach provides a viable solution to significantly increase the overall energy utilisation efficiency, while alleviating the disadvantages of a single approach. A PV thermal collector enables heat harvesting while improving the PV utilisation efficiency by controlling the temperature of PV modules.

Design. Fig. 2 shows the design of a hybrid solar panel utilising PV, thermoelectric and hot-water modules in a multi-layered configuration. The PV surface layer is followed by the thermoelectric layer. The PV cells convert the sun's electromagnetic radiation into electricity, while the thermoelectric layer converts the sun's heat into electricity. The thermoelectric layer is bonded to a plastic lumber plate through a functionally-graded material inter-layer where hot-water tubes are cast. The functionally graded material layer contains aluminium powder dispersed in a high-density polyethylene (HDPE) matrix with a graded microstructure. Water pipelines are cast within the functionally graded material layer to control the panel's temperature.

The plastic lumber made of recycled polymeric materials provides mechanical support and heat insulation of the building skin. The design has the following advantages:

1. The temperature difference between the PV module and the water

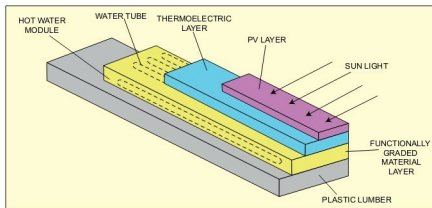


Fig. 2: Hybrid solar panel

tubes provides a considerable temperature gradient within the thermoelectric layer for a higher efficiency of thermoelectric utilisation.

2. The hot-water tubes whose temperature is partially controlled by the flow rate can be directly utilised by water-heating systems for domestic usage.

3. Due to the temperature control on the roof, the room temperature can be significantly reduced and the thermal comfort in the building improved.

4. A high percentage of aluminium powder rapidly transfers heat into the water tubes, but below them the heat conduction is blocked by the HDPE and the plastic lumber substrate.

5. The thin-film PV layer improves heat conduction and structural integrity within the panel and protects the plastic lumber from ultraviolet radiation.

Materials. The PV layer uses commercial single-crystalline silicon solar cells with an open-circuit voltage (Voc) of 0.55V, short-circuit current (Isc) of 4400 mA and energy conversion efficiency of 13 per cent at 25°C under 1000W/m² irradiation. Each panel size is 12.5×12.5 cm² with 270µm thickness.

The thermoelectric layer is made of 97 bismuth-telluride-alloy thermocouples arranged electrically in series and thermally in parallel to form the thermoelectric module measuring 2.9×2.9×0.5 cm³.

Pure aluminium powder and HDPE matrix are used to fabricate the functionally graded material layer. The 6.4mm dia. water tubes are made with copper. The plastic lumber substrate is made of recycled polymeric materials instead of conventional bismuth-telluride modules. If the design is adopted with quantum-well (QW) and quantum-dot (QD) thermoelectric modules, much higher efficiency can be achieved. However, currently most quantum-well and quantum-dot structures are fabricated with molecular beam epitaxial technique, which is expensive.

A promising new technique using colloidal method is now emerging. It can provide quantum-dot thermoelectric structures at a much lower price in the future. To fabricate the functionally graded material layer, the HDPE and aluminium powder are mixed layer by layer with 50 per cent aluminium volume fraction at the top layer and pure HDPE aluminium layer at the bottom.

The moulded functionally graded material and aluminium powder matrix with dimensions of 30.5×30.5×1.5 cm³ is baked in vacuum at 200°C for 1.5 hours. Copper tubes are cast into functionally graded material layer with a separation of 2.5 cm. The functionally graded material layer is bonded to the plastic lumber substrate by vacuum sintering. Thermally conductive paste with a 1.9W/mK conductivity is used as an adhesive between